Math 534H - Spring 2026 - Lecture 10

Some examples:

1.
$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 \\ u(x,0) = \sin x \\ u_t(x,0) = 0 \end{cases}$$

general sol

$$u(x,t) = F(x-ct) + G(x+ct)$$

Using D'Alembert's formula.

$$u(x,t) = \frac{1}{2}\left[\phi(x+ct) + \phi(x-ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)\,dy$$

$\psi(y) = 0$

$\phi(x) = \sin x$

$\psi(x) = 0$

$$u(x,t) = \frac{1}{2}\left[\sin(x+ct) + \sin(x-ct)\right] \Rightarrow$$

$$u(x,t) = \frac{1}{\cancel{2}}\left[\cancel{2}\sin x \cos(ct)\right]$$

$$\underline{u(x,t) = \sin x \cos(ct)} \quad (*)$$

<u>Check</u>: Check IC: $u(x,0) = \sin x \cdot \cos(c \cdot 0)$

$$= \sin x \cdot \underbrace{\cos 0}_{=1}$$

$$= \sin x. \quad \checkmark$$

$$u_t(x,t) = \sin x\,(-\sin(ct))(c)$$

$$= -c \sin x \underbrace{\sin(ct)}_{t=0 \text{ this is } 0} \quad (*)$$

$$u_t(x,0) = 0 \quad \checkmark$$

<u>Check equation</u>: From $(*)$ $u_{tt}(x,t) = -c \sin x \cos(ct)\, c$

$$\underline{u_{tt}(x,t)} = -c^2 \sin x \cos(ct)$$

Next From $(*)$

$$u_x(x,t) = \cos x \cos(ct); \quad \underline{u_{xx}(x,t)} = (-\sin x)\cos(ct)$$

Hence

$$u_{tt}(x,t) - c^2 u_{xx}(x,t) = -c^2 \sin x \cos(ct) + c^2 \sin \cos(ct)$$
$$= 0 \quad \text{as desired.}$$

2

$$\begin{cases} u_{tt} - 16\, u_{xx} = 0 \\ u(x,0) = 0 \\ u_t(x,0) = \cos x \end{cases}$$

From D'Alambert's formula

$$u(x,t) = \frac{1}{8} \int_{x-4t}^{x+4t} \cos y \, dy$$

$c = 4$

So $u(x,t) = \frac{1}{8}\left[\sin(x+4t) - \sin(x-4t)\right]$

$$= \frac{1}{8}\left[2 \cos x \sin(4t)\right]$$

$$= \frac{1}{4} \cos x \sin(4t)$$

CLASS HMWK

Check !

3 $3u_{tt} + u_{xx} - 4u_{xt} = 0$. Is this hyperbolic?

yes

$a_{11} = 1 \quad a_{22} = 3 \quad a_{12} = -2 \qquad a_{22}a_{11} - a_{12}^2 =$

$3 - (-2)^2 = -1 < 0$

Hyperbolic

As we explained in lecture 8 we can complete the squares, change variables and convert this equation to a wave equation in the new variables.

So given IVP:

$$\begin{cases} 3u_{tt} + u_{xx} - 4u_{xt} = 0 \\ u(x,0) = 2x \\ u_t(x,0) = 0 \end{cases}$$

Solve it! Consider $(\partial_x - 2\partial_t)^2 u - \partial_t^2 u$

$$3u_{tt} + u_{xx} - 4u_{xt} = \underbrace{u_{xx} - 4u_{xt} + 4u_{tt}} - 4u_{tt} + 3u_{tt}$$

So we have $\underbrace{(\partial_x - 2\partial_t)}_{\partial_\xi}{}^2 u - \underbrace{\partial_t^2}_{\partial_\eta} u = 0$

Want to change variables so that

$$\boxed{\begin{aligned} \partial_x - 2\partial_t &= \partial_\xi \\ \partial_t &= \partial_\eta \end{aligned}} \quad (\dagger)$$

Because then we have $u_{\xi\xi} - u_{\eta\eta} = 0$ (**)

<u>Remark</u> Note that $(\partial_x - 2\partial_t)^2 u - \partial_t^2 u = 0$

$$\Longleftrightarrow \quad (\partial_x - 2\partial_t - \partial_t)(\partial_x - 2\partial_t + \partial_t) u = 0$$

$$\Longleftrightarrow \quad (\partial_x - 3\partial_t)(\partial_x - \partial_t) u = 0$$

We can guess from what we did before in Lectures 8, 9 that $u(x,t)$ should be $f(3x+t) + g(x+t)$ for f, g general differentiable functions of one variable.

One way would be to check that $u(x,t) = f(3x+t) + g(x+t)$

satisfies $3u_{tt} + u_{xx} - 4u_{xt} = 0$ and then use the I.C. to find what is f and g.

Let's continue from $\circledast\circledast$ (we'll get to some answer).

$$x = \alpha\xi + \beta\eta \qquad \partial_\xi = \partial_x \alpha + \partial_t \cdot \gamma$$
$$t = \gamma\xi + \delta\eta \qquad \partial_\eta = \partial_x \beta + \partial_t \cdot \delta$$

From $\dagger$ we get $\alpha = 1$, $\gamma = -2$, $\beta = 0$, $\delta = 1$

$$\begin{aligned} x &= \xi \\ t &= \eta - 2\xi \end{aligned} \Rightarrow \begin{cases} \xi = x \\ \eta = 2x + t \end{cases}$$

$\therefore$ Our equation in the new variables is $\partial_\xi^2 u - \partial_\eta^2 u = 0$

Hence $u(\xi, \eta) = f(\xi + \eta) + g(\xi - \eta)$ where

f, g are general differentiable functions of one variables. So coming back to (x,t) we get the general solution to $3u_{tt} + u_{xx} - 4u_{xt} = 0$

is $u(x,t) = f(3x+t) + g(-x-t)$

$$\Rightarrow \quad u(x,t) = f(3x+t) + \tilde{g}(x+t) \quad (*)$$

where $f, \tilde{g}$ are general differentiable functions of one variable. Finally we need to determine what is f and what is $\tilde{g}$ so that

$$\begin{cases} u(x,0) = 2x \\ u_t(x,0) = 0 \end{cases} \quad (**)$$

$$2x \overset{(**)}{=} u(x,0) \overset{(*)}{=} f(3x) + \tilde{g}(x) \quad (1)$$

Next compute $u_t(x,t) = f'(3x+t).1 + \tilde{g}'(x+t).1 \Rightarrow$

Hence

$$0 \overset{**}{=} u_t(x,0) = f'(3x) + \tilde{g}'(x) \quad \Big] \; 2$$

We get from 1 and 2 that

$$\begin{cases} 2x = f(3x) + \tilde{g}(x) \\ 0 = f'(3x) + \tilde{g}'(x) \end{cases}$$

We need to find f and $\tilde{g}$

To do this by differentiating the first equation in x we get

$$\begin{cases} 2 = 3f'(3x) + \tilde{g}'(x) \\ 0 = f'(3x) + \tilde{g}'(x) \end{cases}$$

Now solve for f' and $\tilde{g}'$

we get that $2 = 2f'(3x) \Rightarrow f'(3x) = 1$

But since this is true $\forall x$ by calling

$z = 3x$, we have $f'(z) = 1 \Rightarrow f(z) = z + c_1$

Multiply second eq. by -3 and add the two to get

$$2 = -2\tilde{g}'(x) \Rightarrow \tilde{g}'(x) = -1$$

$$\Rightarrow \tilde{g}(w) = -w + c_2$$

Hence $u(x,t) = \underbrace{(3x+t)+c_1}_{f(3x+t)} + \underbrace{(-x-t)+c_2}_{\tilde{g}(x+t)}$

$u(x,t) = 2x + c_1 + c_2$. But now recall <u>that</u>

$2x = u(x,0) \Longrightarrow c_1 + c_2 = 0.$

Initial position

All in all we get $\boxed{u(x,t) = 2x}$ solves

$$\begin{cases} 3u_{tt} + u_{xx} - 4u_{xt} = 0 \\ u(x,0) = 2x \\ u_t(x,0) = 0. \end{cases}$$